

# PERFORMANCE EVALUATION OF CLASSIFICATION MODELS

## FOR CREDIT CARD FRAUD DETECTION

A Project Report Submitted During Internship

|                                 |                                                               |
|---------------------------------|---------------------------------------------------------------|
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# Introduction

The global shift toward convenient digital payments has driven a rise in credit card fraud, making automated detection systems an essential requirement for modern financial security. This project investigates the application of machine learning techniques for detecting fraudulent credit card transactions using a dataset containing **100,000** transactions. A notable characteristic of the dataset is the presence of only **1% fraudulent transactions**. Therefore, greater emphasis is placed on evaluation metrics such as Precision, Recall, F1-Score, ROC-AUC, and Precision–Recall Curves rather than overall accuracy.

## Methodology

The analysis follows a four-stage framework:

1. **Exploratory Data Analysis (EDA):** Examination of transaction characteristics, fraud distribution, and feature relationships.
2. **Feature Engineering:** Extraction of temporal variables such as Month, Day of Week, and Weekend indicators from transaction dates.
3. **Model Development and Comparison:** Implementation of Logistic Regression, Decision Tree, Random Forest, and XGBoost classifiers.
4. **Operational Evaluation:** Assessment of model performance using confusion matrices, ROC curves, Precision–Recall curves, threshold tuning, calibration analysis, and cost analysis.

## Overview of the Dataset

The original dataset contains 7 variables. Additional temporal features such as Month, DayOfWeek, and Weekend were derived during feature engineering. Only 1% of transactions are fraudulent, so class imbalance is severe. Key features include:

- **TransactionID:** unique transaction identifier (dropped before modeling)
- **TransactionDate:** timestamp of the transaction
- **Amount:** monetary value
- **MerchantID:** identifier for the merchant (dropped)
- **TransactionType:** Purchase or Refund
- **Location:** city of transaction
- **IsFraud:** target label (1=fraud, 0=legitimate)

There are no missing values. 1,000 transactions are frauds and 99,000 are normal. We also derive Month, DayOfWeek, and Weekend from the date to capture temporal patterns.

## Sample Records from Dataset

- Total Records: 100,000
- Total Features: 7
- Fraud Transactions: **1,000**
- Legitimate Transactions: **99,000**
- Fraud Rate: **1%**
- No of Missing values: 0

# First Five Records of the Dataset

| TransactionID | TransactionDate | Amount  | MerchantID | TransactionType | Location     | IsFraud |
|---------------|-----------------|---------|------------|-----------------|--------------|---------|
| 1             | 03-04-2024      | 4189.27 | 688        | refund          | San Antonio  | 0       |
| 2             | 19-03-2024      | 2659.71 | 109        | refund          | Dallas       | 0       |
| 3             | 08-01-2024      | 784.00  | 394        | purchase        | New York     | 0       |
| 4             | 13-04-2024      | 3514.40 | 944        | purchase        | Philadelphia | 0       |
| 5             | 12-07-2024      | 369.07  | 475        | purchase        | Phoenix      | 0       |

## Data Preprocessing

- Converted transaction dates into datetime format
- Extracted Month, DayOfWeek, and Weekend features
- Removed TransactionID and MerchantID
- Applied one-hot encoding to categorical variables
- Performed train-test split (80%-20%)

## Exploratory Data Analysis

Fraud rates for refund (1.01%) and purchase (0.99%) transactions are almost identical. Therefore, transaction type shows little association with fraud occurrence and is unlikely to contribute significantly to fraud prediction.

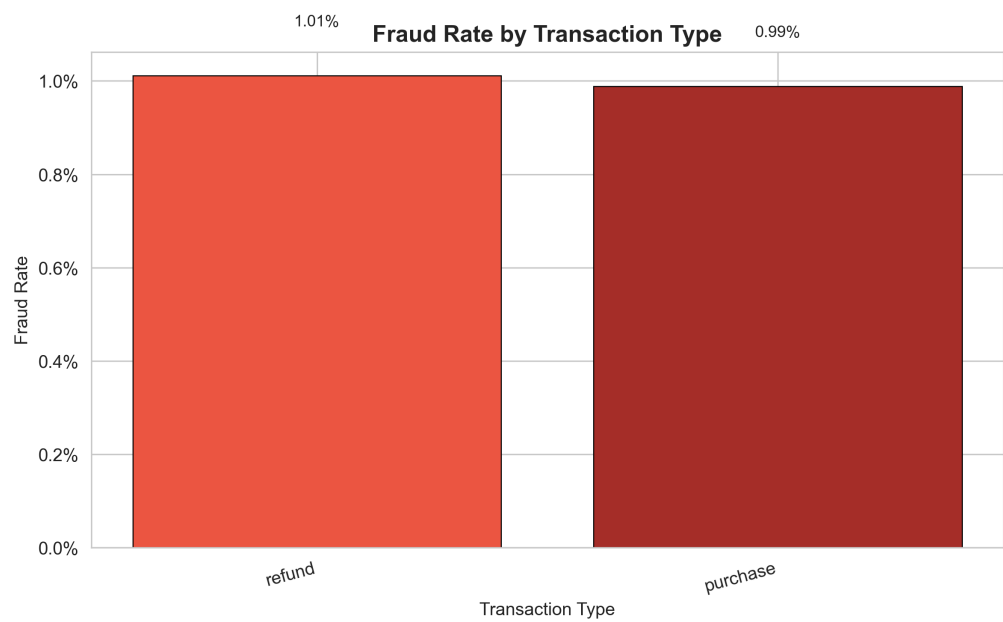


Figure 1: Fraud Rate by Transaction Type

Fraud rates across locations lie within a narrow range of 0.89%–1.16%. The observed differences are modest, suggesting that location has limited influence on fraud occurrence in this dataset.

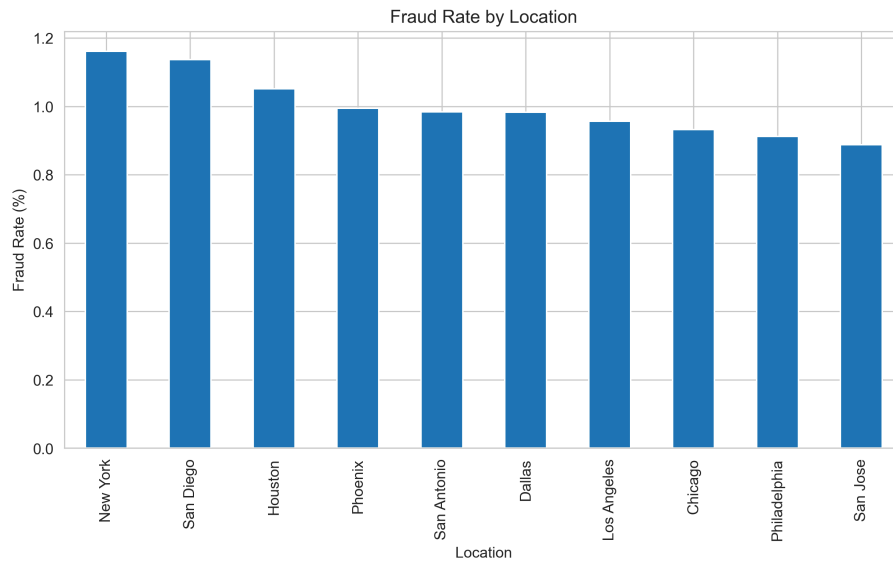


Figure 2: Fraud Rate by Locations

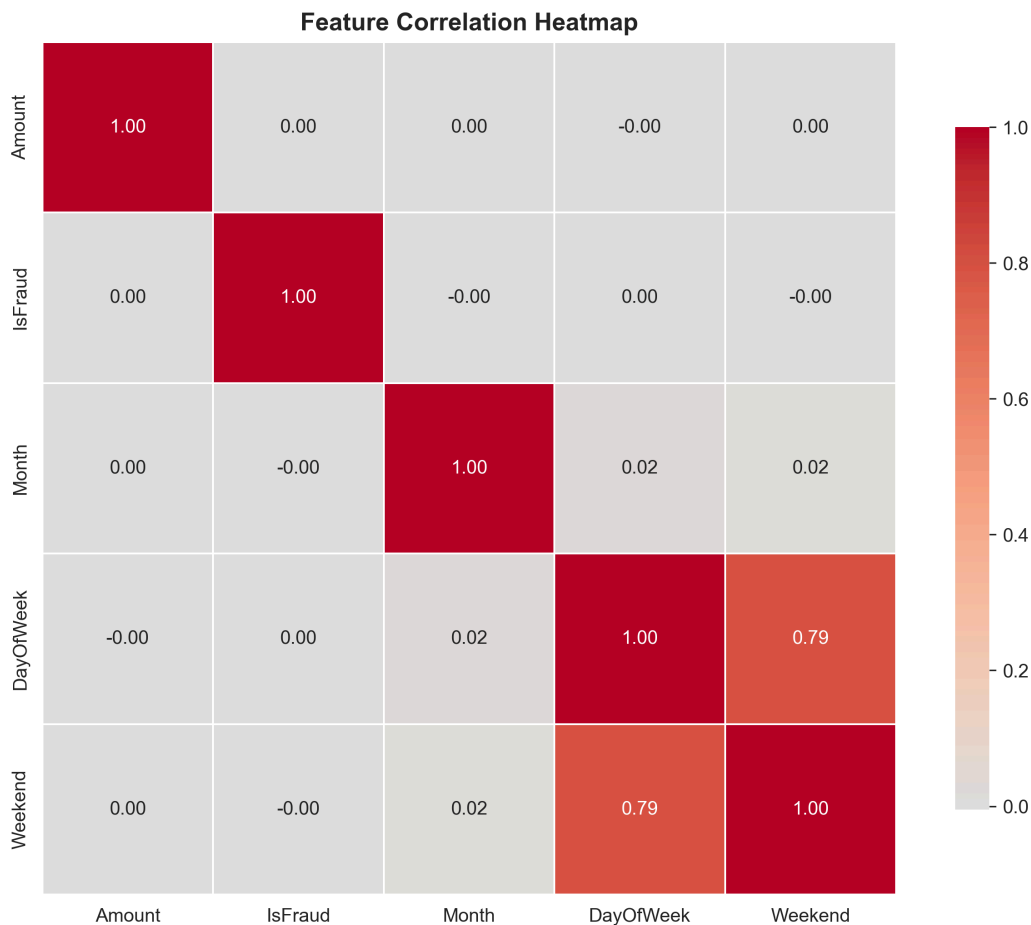


Figure 3: Transaction Amount Distribution by Fraud Status

The boxplots for fraudulent and non-fraudulent transactions are highly similar. Both groups exhibit comparable medians, interquartile ranges, and overall spread of transaction amounts. No prominent outliers are observed in either group. This suggests that transaction amount alone does not provide strong discrimination between fraudulent and legitimate transactions in the dataset.

## Feature Correlation

A Pearson correlation heatmap shows essentially zero correlation ( $|r| < 0.01$ ) between **IsFraud** and any feature (Amount, Month, DayOfWeek, Weekend). The only strong correlation is between **DayOfWeek** and **Weekend** (by construction). Thus, No feature shows a strong linear relationship with fraud.



## Modeling

We trained the following classifiers (with key hyperparameters):

- Logistic Regression:
- Decision Tree:
- Random Forest:
- XGBoost:

All models were fit on an 80/20 stratified train/test split, with class weighting to compensate for 1% fraud.

## Evaluation Metrics

The following evaluation metrics are used to assess the performance of the classification models:

- **Accuracy:**  $(TP + TN) / (TP + TN + FP + FN)$ , representing the overall proportion of correctly classified transactions.
- **Precision:**  $TP / (TP + FP)$ , representing the proportion of transactions predicted as fraud that are actually fraud.
- **Recall (Detection Rate):**  $TP / (TP + FN)$ , representing the proportion of actual fraud transactions successfully detected.

- **Specificity:**  $TN/(TN + FP)$ , representing the proportion of legitimate transactions correctly identified as non-fraud.
- **F1-Score:** Harmonic mean of Precision and Recall, providing a balance between fraud detection and false alarms.
- **ROC-AUC:** Area under the Receiver Operating Characteristic curve, measuring the model's ability to distinguish between fraud and legitimate transactions across different thresholds.
- **PR-AUC (Average Precision):** Area under the Precision-Recall curve. This metric is particularly important for highly imbalanced datasets because it focuses on the detection of the minority (fraud) class.
- **Confusion Matrix:** Summarizes True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN), providing a detailed view of classification errors.

Since the dataset contains only about **1% fraud transactions**, accuracy alone may be misleading. For example, a model that predicts every transaction as legitimate would achieve approximately 99% accuracy while completely failing to detect fraud. Therefore, Precision, Recall, F1-Score, ROC-AUC, and PR-AUC are more informative measures of model performance in this study.

Threshold tuning is also explored. Lowering the decision threshold generally increases Recall by identifying more fraud transactions, but it often reduces Precision by generating more false alerts. This trade-off is important in practical fraud detection systems.

## ROC Curve

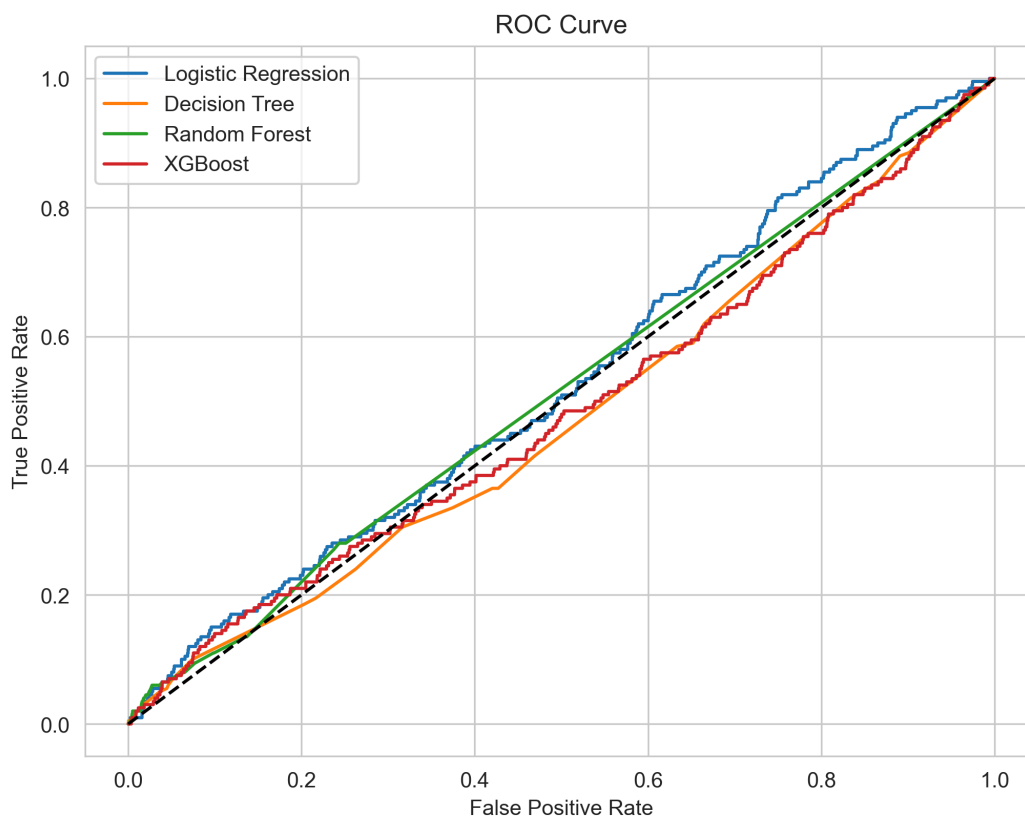


Figure 5: ROC Curve Comparison of Classification Models

The ROC curves lie close to the diagonal reference line. This indicates that all four models have limited ability to distinguish fraudulent transactions from legitimate ones. Logistic Regression shows slightly better performance than the other models, but the overall differences are small. The similar performance of all four models indicates that the available variables provide only weak information for separating fraudulent and legitimate transactions.

## Precision–Recall Curve

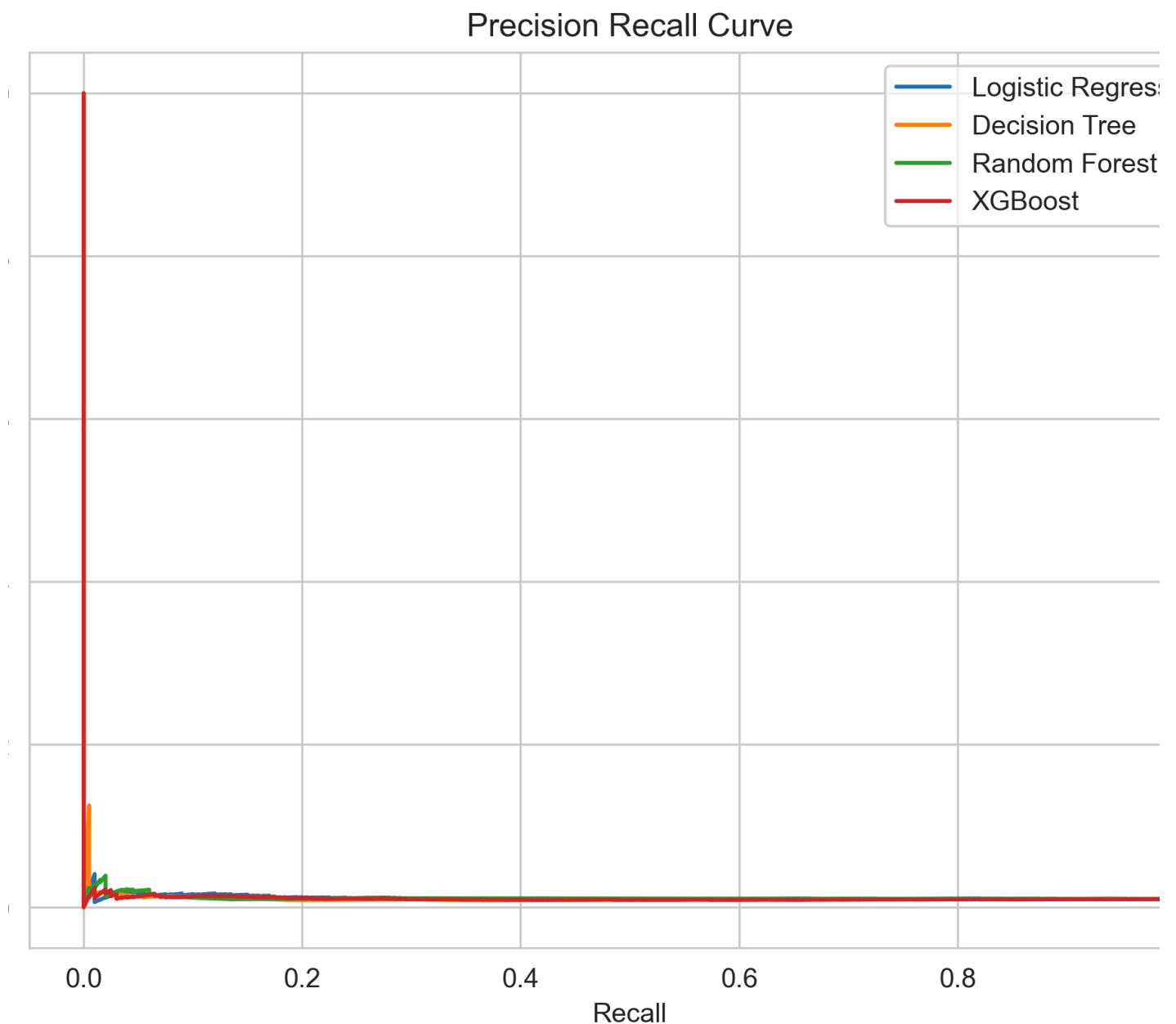


Figure 6: Precision–Recall Curve Comparison of Classification Models

The Precision–Recall curves remain near the baseline for almost the entire range of recall values. This reflects the highly imbalanced nature of the dataset, where fraud transactions account for only about 1% of all observations. None of the models achieves a useful balance between precision and recall, indicating that fraud patterns are weakly represented in the available features.

# Confusion Matrices

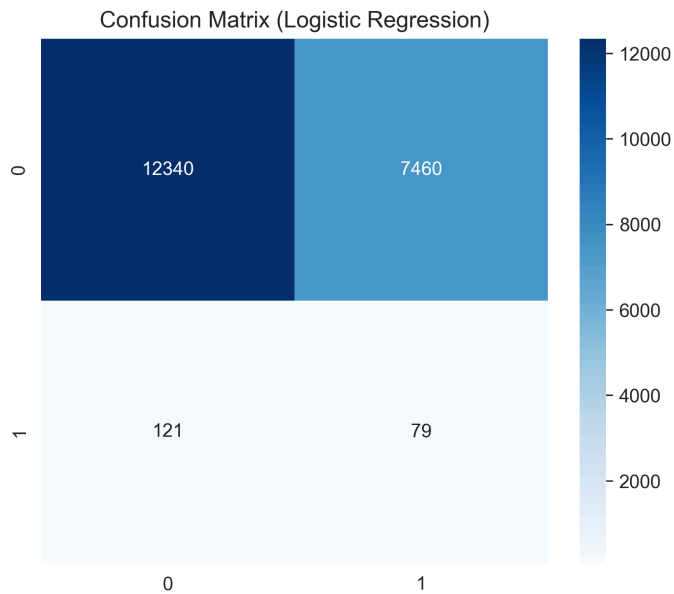


Figure 7: Logistic Regression

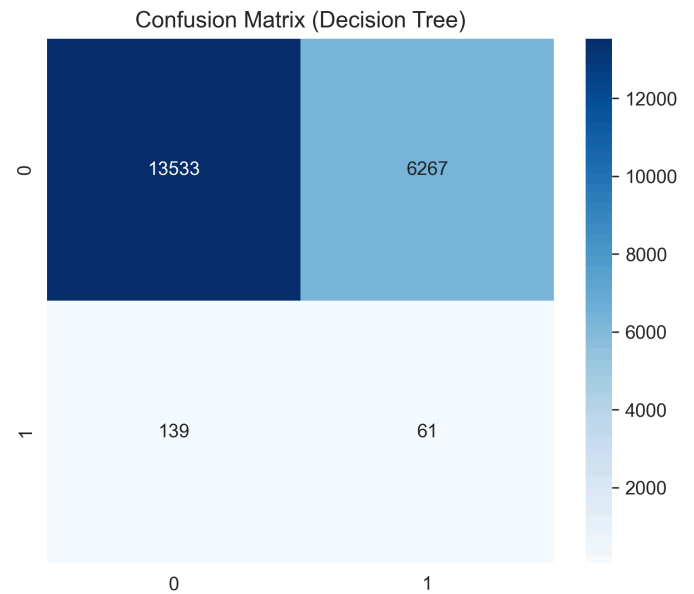


Figure 8: Decision Tree

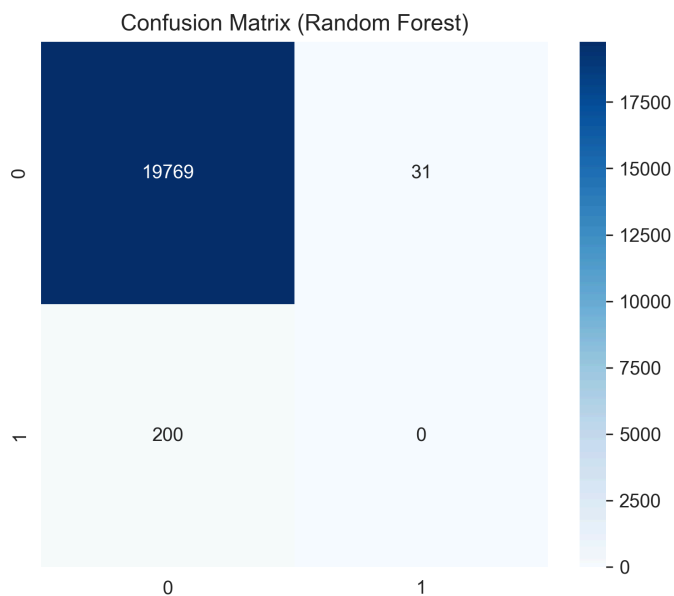


Figure 9: Random Forest

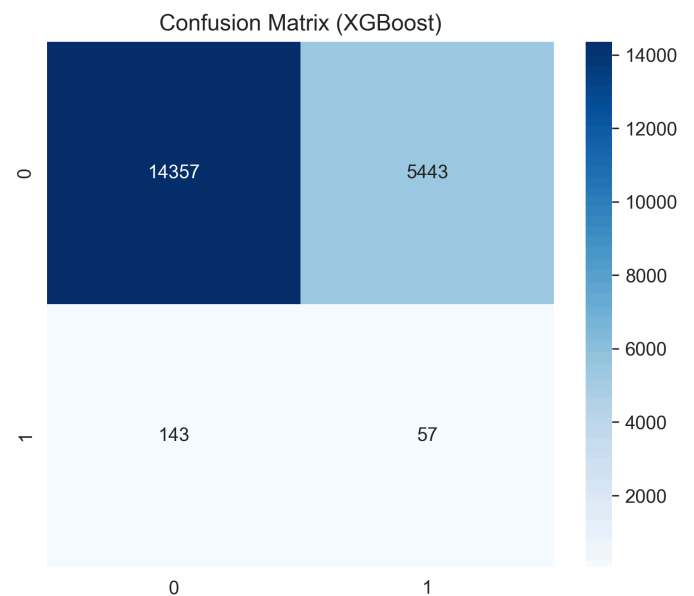


Figure 10: XGBoost

- Logistic Regression identified the largest number of fraud transactions (79 true positives) but also generated a large number of false positives (7,460).
- Decision Tree and XGBoost showed similar performance, correctly identifying 61 and 57 fraud transactions, respectively.
- Random Forest produced the fewest false positives (31) but failed to identify any fraud transaction, classifying all fraud cases as legitimate transactions.
- Overall, none of the models demonstrated strong fraud detection capability, indicating that the available features contain limited information for distinguishing fraud from non-fraud transactions.



## Feature Importance

Feature importance measures the relative contribution of each predictor variable to the Random Forest model. Variables with higher importance values have a greater influence on the model's classification decisions, while variables with lower values contribute little to the prediction process.

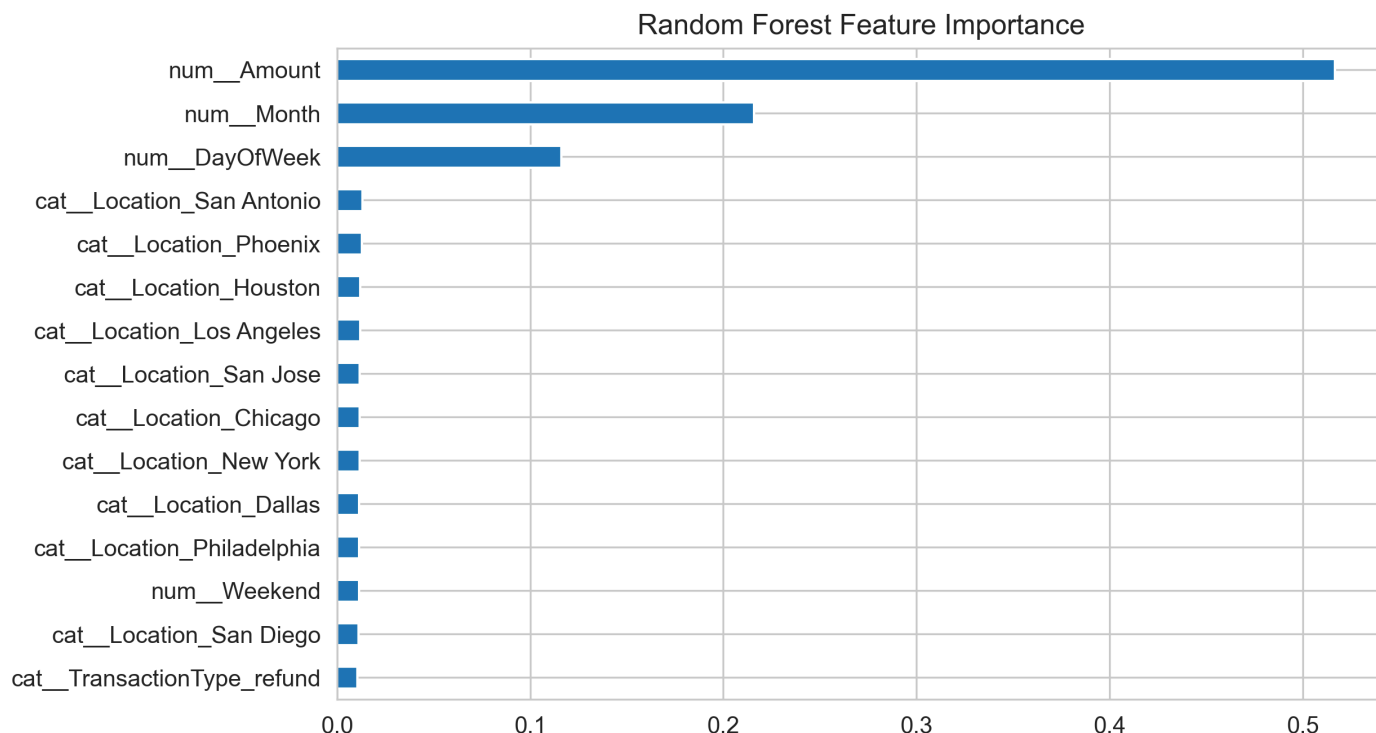


Figure 11: Random Forest Feature Importance

- Transaction Amount was the most influential variable, accounting for more than half of the total feature importance. This indicates that the model relied primarily on transaction amount when making classification decisions.
- Month and Day of Week were the next most important variables, although their contributions were considerably smaller than that of transaction amount.
- Location-related variables and Transaction Type contributed very little to the model. Their importance values were close to zero, suggesting that these variables provided limited information for distinguishing fraud from legitimate transactions.
- The dominance of only a few variables, combined with the weak predictive performance observed earlier, suggests that the available features do not contain strong signals for fraud detection.

# Threshold Tuning

Threshold tuning is the process of selecting an optimal probability cutoff for classifying observations into fraud and non-fraud classes. By default, most classification models use a threshold value of 0.5. However, for highly imbalanced datasets such as credit card fraud detection, adjusting the threshold can improve the balance between precision, recall, and F1-score. The objective is to identify a threshold that maximizes fraud detection performance while controlling false positive predictions.

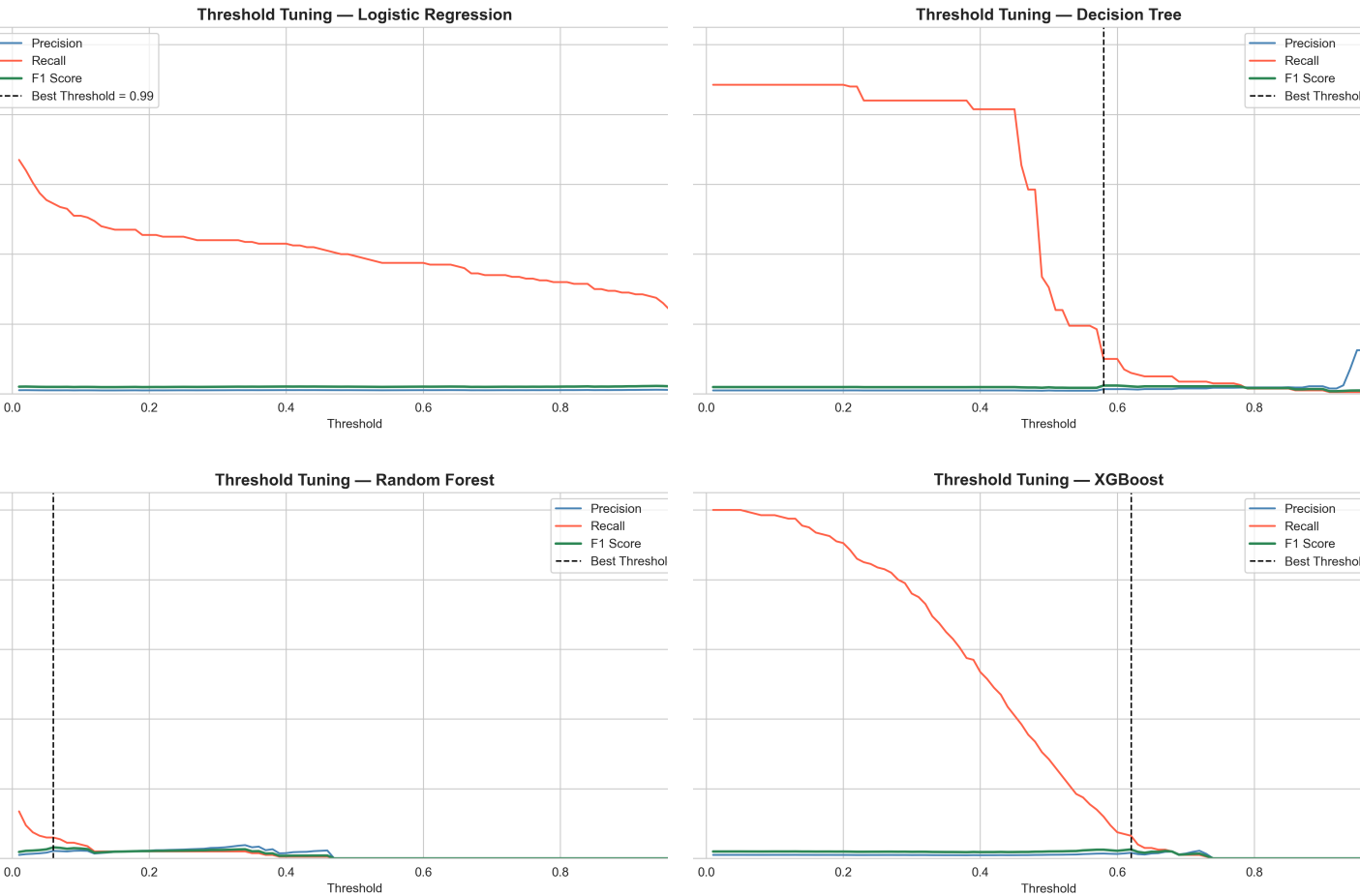


Figure 12: Threshold Tuning Curves for Different Classification Models

| Model               | Best Threshold | Precision | Recall | F1-Score |
|---------------------|----------------|-----------|--------|----------|
| Logistic Regression | 0.99           | 0.0122    | 0.190  | 0.0230   |
| Decision Tree       | 0.58           | 0.0135    | 0.100  | 0.0237   |
| Random Forest       | 0.06           | 0.0215    | 0.060  | 0.0317   |
| XGBoost             | 0.62           | 0.0161    | 0.065  | 0.0258   |

Table 1: Optimal Threshold and Performance Metrics for Different Models

- Logistic Regression achieved its best performance at a threshold of 0.99. At this threshold, the model detected 19% of the fraud cases present in the test data, which was the highest recall among all models. However, its precision remained very low (1.22%), indicating that many legitimate transactions were incorrectly classified as fraud.
- Decision Tree performed best at a threshold of 0.58. The model showed slightly higher precision than Logistic Regression but identified only 10% of fraud cases. Its F1-score remained low, suggesting limited overall effectiveness in balancing fraud detection and false alarms.

- Random Forest selected a much lower optimal threshold of 0.06 and achieved the highest F1-score (0.0317) among the four models. Nevertheless, the improvement was marginal, as the model detected only 6% of fraud cases and maintained a low precision value.
- XGBoost achieved its best performance at a threshold of 0.62. The model detected 6.5% of fraud cases with a precision of 1.61%. Although its performance was comparable to the other models, it did not provide any significant advantage in identifying fraudulent transactions.
- Overall, threshold tuning led to only minor improvements in model performance. Even after selecting the optimal threshold for each model, the precision, recall, and F1-scores remained very low. This suggests that the available features in the dataset have weak predictive power for fraud detection, making it difficult for the models to effectively separate fraudulent and legitimate transactions.

## Calibration Curve Analysis

A calibration curve evaluates how well a model’s predicted probabilities correspond to the actual observed outcomes. A perfectly calibrated model produces probability estimates that closely match the true frequency of fraud and follows the diagonal reference line. Large deviations from the diagonal indicate that the predicted probabilities are not reliable representations of the actual fraud risk.

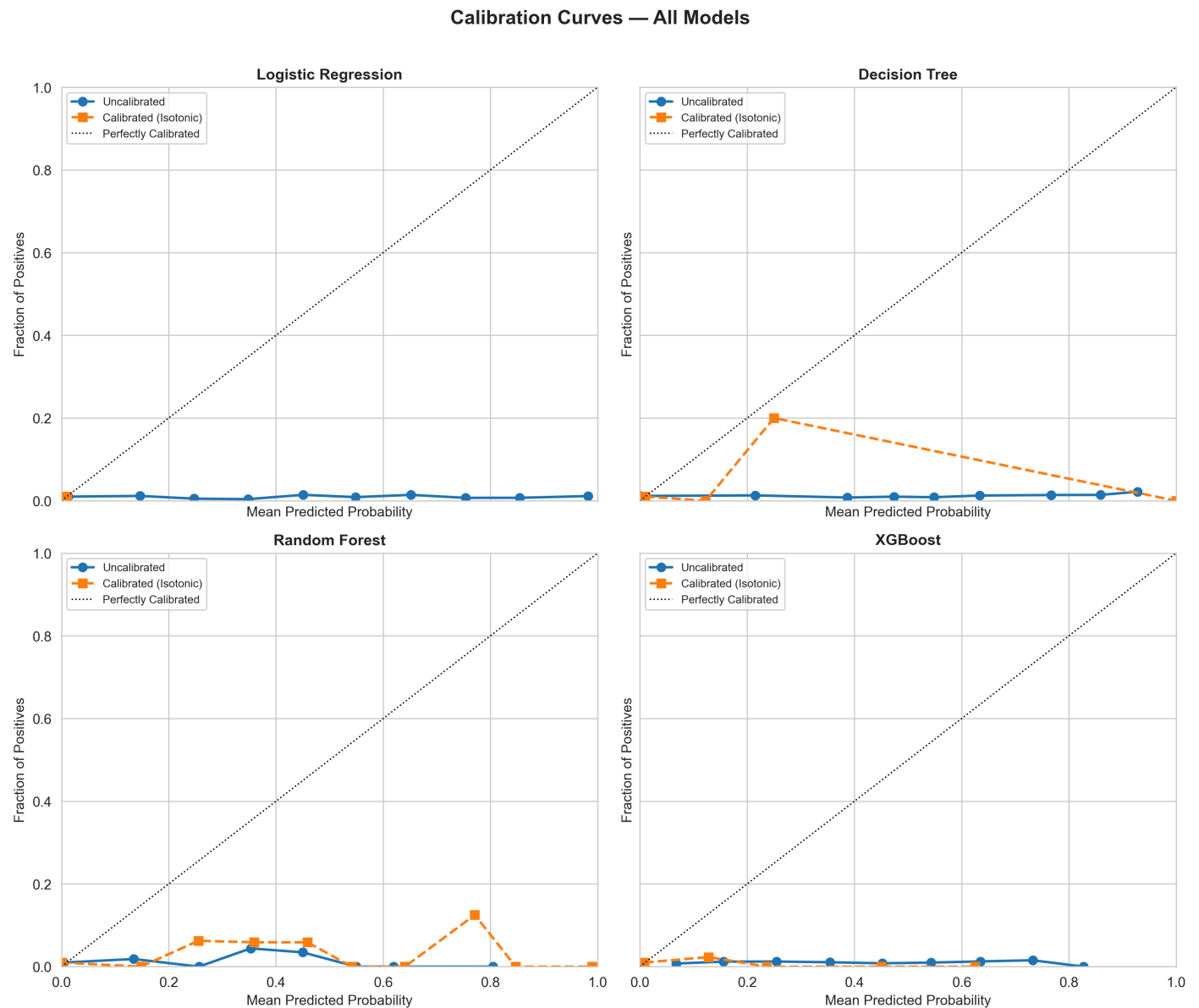


Figure 13: Calibration Curves for Logistic Regression, Decision Tree, Random Forest, and XGBoost

- The calibration curves of all four models deviate substantially from the ideal diagonal line, indicating poor probability calibration.
- Logistic Regression and XGBoost produce probability estimates that remain concentrated near very low values, reflecting the severe class imbalance present in the dataset.
- Decision Tree and Random Forest show greater fluctuations after calibration, but the calibrated curves still fail to closely follow the ideal reference line.
- The results suggest that the predicted probabilities cannot be interpreted as reliable estimates of fraud risk. Even after isotonic calibration, the models struggle to distinguish fraud transactions from legitimate transactions.
- Overall, the calibration analysis supports the findings from the ROC, Precision-Recall, and confusion matrix results: the available features provide limited information for effective fraud prediction, resulting in weak classification performance across all models.

The calibration results indicate that none of the models produce well-calibrated probability estimates, further confirming the difficulty of fraud detection using the available variables in the dataset.

## Cost-Based Threshold Analysis

Traditional evaluation metrics such as accuracy, precision, recall, and F1-score treat all prediction errors equally. In real-world fraud detection, however, different mistakes have very different financial consequences.

A **False Negative (FN)** occurs when an actual fraud transaction is predicted as legitimate. This is usually the most expensive error because the fraud passes through the system and results in a direct financial loss.

A **False Positive (FP)** occurs when a legitimate transaction is flagged as fraud. Although inconvenient, this error generally causes a much smaller loss, such as additional verification costs, operational workload, or customer dissatisfaction.

For this analysis, the following cost assumptions were used:

- Cost of one missed fraud (False Negative) = £500
- Cost of one false alert (False Positive) = £10

The total cost was calculated as:

$$\text{Total Cost} = (\text{FN} \times £500) + (\text{FP} \times £10)$$

Different classification thresholds were tested from 0 to 1. For each threshold, the confusion matrix was computed and the corresponding total cost was calculated. The threshold producing the minimum total cost was considered the optimal threshold for that model.

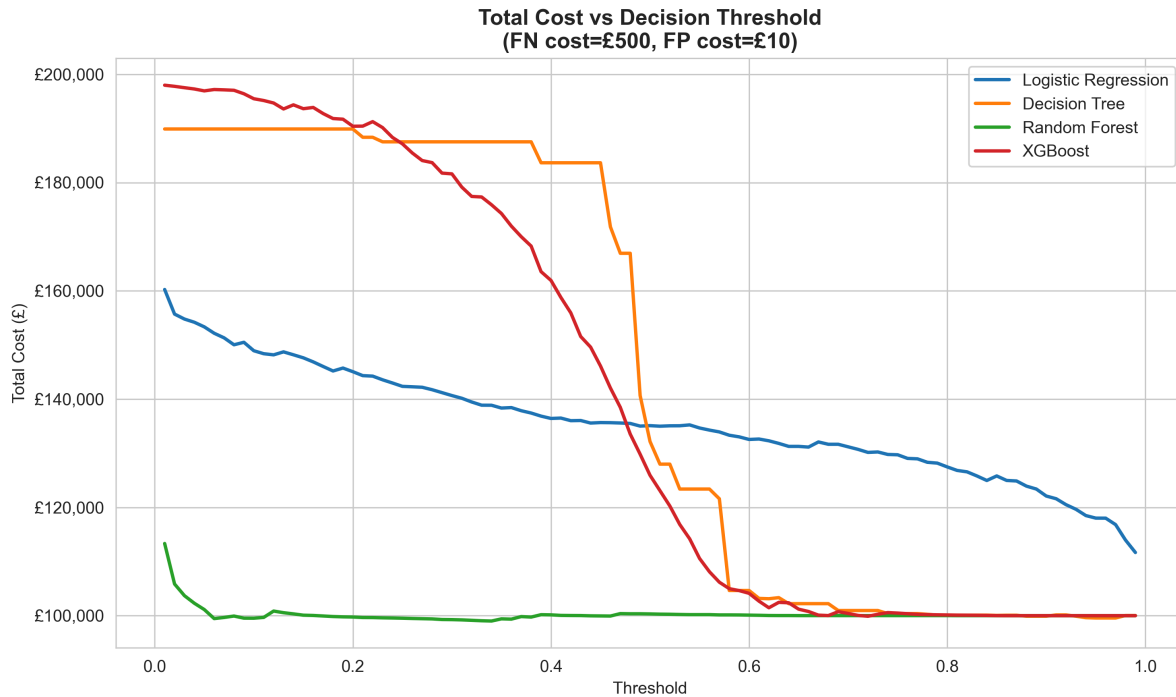


Figure 14: Total Cost versus Classification Threshold for Different Models

- The figure shows how the total financial cost changes as the classification threshold varies for each model. Lower curves indicate better performance because they correspond to lower overall financial loss.
- Logistic Regression shows a gradual reduction in cost as the threshold increases, eventually reaching its lowest cost near the higher threshold region. This suggests that stricter fraud classification reduces unnecessary false alerts while maintaining reasonable fraud detection.
- Decision Tree and XGBoost begin with very high total costs because a large number of fraud transactions are missed at lower thresholds. Their costs decrease sharply around the middle threshold range, where the balance between missed fraud and false alerts improves.
- Random Forest consistently produces the lowest overall cost across most threshold values. Although its fraud detection performance was weak according to recall and F1-score, the model generated very few false alerts, resulting in a lower total financial cost under the chosen cost assumptions.
- Around threshold values greater than 0.6, the costs of Decision Tree, Random Forest, and XGBoost become very similar, approaching approximately £100,000. This indicates that beyond this region, changing the threshold provides only limited additional cost reduction.
- The analysis highlights an important practical point: the model with the best statistical metrics is not always the model with the lowest financial cost. Cost-sensitive evaluation focuses on real-world consequences rather than classification accuracy alone.

To account for the different costs associated with missed fraud cases and false alerts, threshold tuning was performed for each model. The optimal threshold was defined as the classification threshold that minimized the total cost.

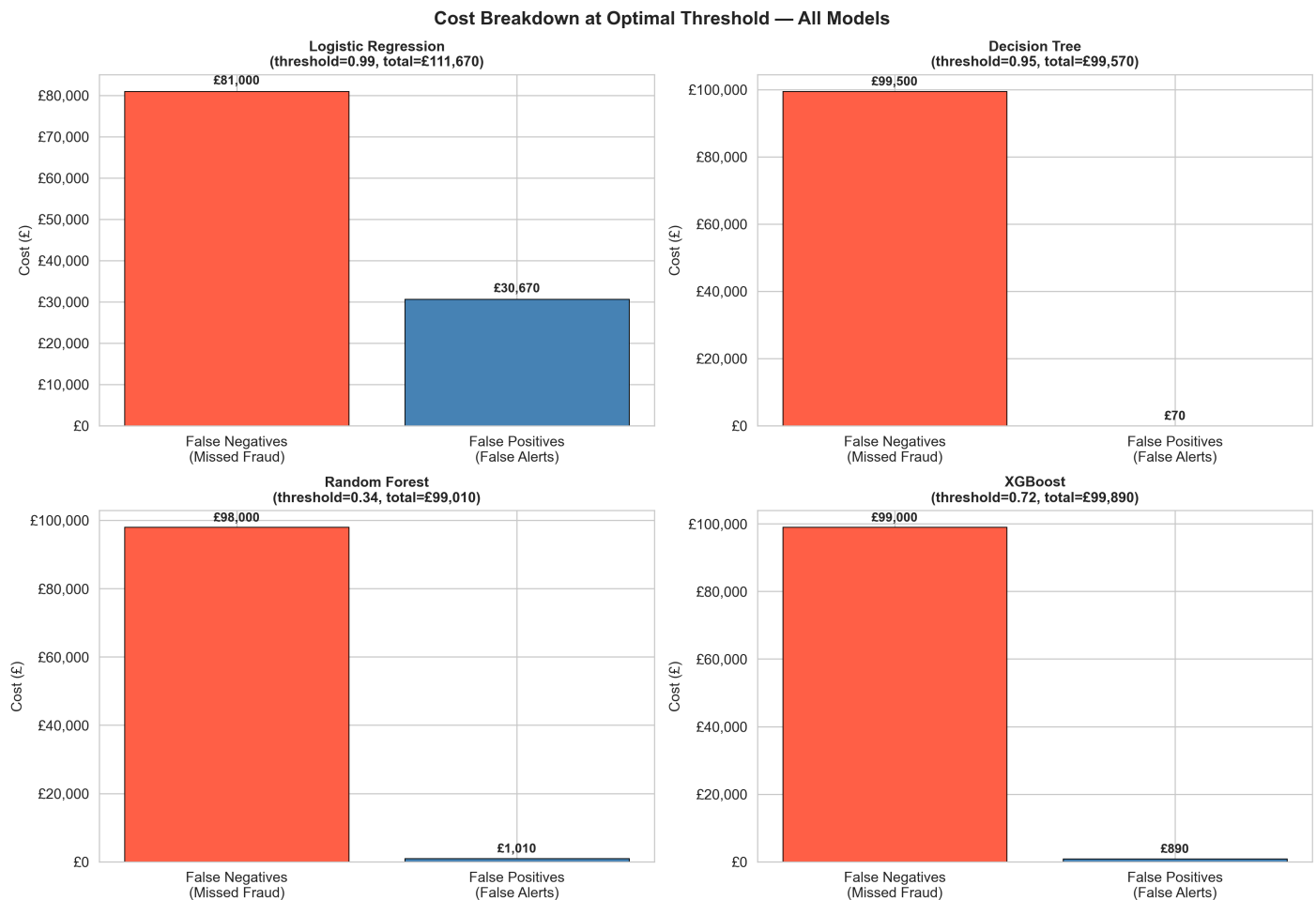


Figure 15: Cost Breakdown at Optimal Threshold for Different Models

- For all models, the majority of the total cost arises from false negatives (missed fraud cases) rather than false positives. This is expected because the cost assigned to a missed fraud (£500) is substantially higher than the cost of a false alert (£10).
- Logistic Regression produced the highest false-positive cost (£30,670), indicating that many legitimate transactions were incorrectly flagged as fraud. However, it incurred a lower false-negative cost (£81,000) than the other models.
- Decision Tree, Random Forest, and XGBoost generated very small false-positive costs, suggesting that they rarely classified legitimate transactions as fraudulent. Nevertheless, these models missed a larger number of fraud cases, causing false negatives to dominate the total cost.
- The minimum total costs obtained by Decision Tree (£99,570), Random Forest (£99,010), and XGBoost (£99,890) were very similar. This indicates that differences between the models had only a limited effect on overall financial loss, as the available features provide weak discriminatory power for fraud detection.

## Conclusion

- This study analyzed a credit card transaction dataset containing information on transaction amount, transaction type, location, and transaction date, with the objective of identifying fraudulent transactions.
- Exploratory data analysis showed that fraud constituted only about **1%** of all transactions, indicating a highly imbalanced classification problem. Fraud rates were also found to be relatively similar across transaction types and locations.
- Correlation analysis revealed very weak relationships between the available predictor variables and the fraud indicator. Transaction amount, month, and day of week exhibited some influence, while location and transaction type contributed little to fraud identification.
- Four classification models—Logistic Regression, Decision Tree, Random Forest, and XGBoost—were trained and evaluated using stratified train-test splitting. Performance assessment through ROC curves, Precision-Recall curves, confusion matrices, and threshold tuning indicated limited predictive capability across all models.
- Logistic Regression detected the largest number of fraud cases but generated a substantial number of false alerts. Random Forest produced the fewest false alerts but failed to identify most fraud cases. Decision Tree and XGBoost achieved similar results, with no model demonstrating a clear advantage.
- Threshold tuning and probability calibration were performed to improve model performance. However, only marginal improvements were observed, and precision, recall, and F1-scores remained low for all models.
- Cost analysis further showed that missed fraud cases contributed the majority of the total financial loss, even at the optimal classification thresholds. The minimum costs achieved by different models were very similar, indicating that model choice had only a limited effect on overall performance.
- Overall, the results suggest that the available variables do not contain sufficient information to effectively distinguish fraudulent transactions from legitimate ones. Improved fraud detection would likely require additional customer-level, behavioral, temporal, or transaction-history features that are not present in the current dataset.